

333 Summer 2026 Final

Name: _____

The exam starts at 9.30am and ends at 11.00 am

1. [20 Points] Sketch the design and pseudocode in Verilog implementation for a L1 instruction cache. Input is the PC bits for the index output is the 32 bits instruction or miss. The L1 cache is a **direct-mapped** cache with **X blocks** and **Y words per block**, using **byte addressing**. ON miss it will call a FSM in charge of getting the data from the next level of the cache, you don't need to create this FSM assume it does exist

2. [10 Points] Assume you have a cache with **128 total blocks**, organized as **8-way set associative**, with **2 words per block**, and using **byte addressing**. Given a **30-bit address**, determine the following:
- How many index bits** are required to select a set?
 - How many word-offset bits** are needed?
 - How many byte-offset bits** are needed?
 - How many tag bits** does the address include?
3. [10 points] Given a **32-bit virtual address space**, where virtual addresses are translated into **29-bit physical addresses**, and the **page size is 8 KB**, determine the **size of the page table** (i.e., *number of entries* $\times$ *size of each entry*).

Be sure to **show all calculations**, as answers without work will receive no credit.

4. [10points] Draw a diagram that illustrates how the TLB, page table, and cache hierarchy interact during address translation and memory access.

5. Assume you have a **direct-mapped L1 cache** with **16 blocks**, each containing **4 words per block**, and the system uses **byte addressing**. If an access misses in the L1 cache, the next level is **main memory**, which has a **12-bit physical address space**.

[20 points] Determine and **SHOW** the **final contents of the L1 cache** after the CPU performs the following memory operations.

The image provided shows how data is loaded into the L1 cache after each access.

For each access, fill in the **Index**, **Tag**, and indicate whether it results in a **Hit or Miss**:

1. **Read(4)** → Index = _____, Tag = _____, Hit or Miss = _____
2. **Read(20)** → Index = _____, Tag = _____, Hit or Miss = _____
3. **Write(0) value 70** → Index = _____, Tag = _____, Hit or Miss = _____
4. **Read(8)** → Index = _____, Tag = _____, Hit or Miss = _____

Note: Numbers in parentheses represent addresses in decimal.

b. [10 points] What is the %Hits ? **Show your work** and how the L1 looks like at the end of the above address

6. **[20 points]** In a **5-stage pipeline with data forwarding**, determine **how many clock cycles** are required to execute the following sequence of instructions:

```
beq    r1, r2, label
lw     r1, 8(r1)
sub    r2, r1, r1
or     r2, r2, r1
sw     r1, 8(r3)
j      exit
label: lw r1, 12(r2)
exit:
```

Assume the following:

- The **branch predictor assumes the branch is not taken**.
- The **branch target address is computed in the EX stage**, and the correct PC (if the branch was taken) is loaded in the MEM stage.
- Analyze the scenario where the **branch predictor is wrong**, and the branch **must be taken**.

Important: Full credit requires **pipeline diagrams** showing all **forwarding paths, stalls, and flushed instructions**.